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HAN Competition Challenge

Overview

The Human-Agent Negotiation (HAN) competition is a bilateral negotiation challenge where two parties (e.g., buyer and seller) negotiate over a finite set of discrete outcomes. Each party has private preferences encoded as a utility function. The goal is to reach a mutually acceptable agreement that maximizes one's own utility.

Outcome Space

Let there be n issues $I_1, I_2, \dots, I_n$, each with a finite discrete domain D_i. The **outcome space** is the Cartesian product:

$$\Omega = D_1 \times D_2 \times \cdots \times D_n$$

An outcome $\omega \in \Omega$ is a tuple $\omega = (v_1, v_2, \dots, v_n)$ where $v_i \in D_i$.

Linear Additive Utility Function

Each negotiator j has a **Linear Additive Utility Function** defined by:

- A weight vector $\alpha^j = (\alpha_1^j, \alpha_2^j, \dots, \alpha_n^j)$ with $\alpha_i^j \geq 0$
- A per-issue value function $f_i^j : D_i \to [0, 1]$ (a discrete mapping, called a **TableFun**)
- A reserved value $r^j \in \mathbb{R}$ (the utility of disagreement)

The utility of an outcome $\omega = (v_1, \dots, v_n)$ for negotiator j is:

$$U^j(\omega) = \sum_{i=1}^{n} \alpha_i^j \cdot f_i^j(v_i)$$

The negotiator prefers agreement on ω over disagreement if and only if $U^j(\omega) > r^j$.

Example (Trade Domain)

Two issues: Quantity ($D_1 = \{1, \dots, 10\}$) and Price ($D_2 = \{100, 110, 120, 200\}$).

Seller: $\alpha_1 = 0.5,\ \alpha_2 = 0.5,\ r = 0.0$

Quantity	$f_1(v)$	Price	$f_2(v)$
1	0.0	100	0.0
5	1.0	110	0.1
6	0.9	120	0.2
10	0.0	200	1.0

For outcome $(5, 200)$: $U^{\text{seller}}(5, 200) = 0.5 \cdot 1.0 + 0.5 \cdot 1.0 = 1.0$

Example (Island Survival Domain)

Two negotiators are stranded on an island with 8 items and must negotiate how to divide them. Each item has some value for each negotiator but neither knows how much their partner values each item.

Eight issues: $D_i = \{\text{alice}, \text{bob}\}$ for each item. Value functions are binary: $f_i^j(\text{self}) = 1,\ f_i^j(\text{other}) = 0$.

Alice: $\alpha = (13, 22, 17, 6, 5, 20, 7, 10),\ r = 0.0$

Bob: $\alpha = (5, 20, 7, 13, 10, 22, 6, 17),\ r = 0.0$

The parties have different priorities over items — negotiation is needed to find a mutually beneficial split.

Example (Grocery Domain)

Two negotiators bought 4 items of 4 types of fruits and are negotiating about how to distribute them between themselves. Each fruit has some value for each negotiator; the more items one gets, the higher utility one receives.

Four issues: $D_i = \{0, 1, 2, 3, 4\}$ (number of each fruit one negotiator gets). One party uses affine value functions $f_i(v) = 1 - 0.25v$ and the other uses linear value functions $f_i(v) = 0.25v$.

Alice: $\alpha = (0.48, 0.32, 0.16, 0.04),\ r = 0.0$

Bob: $\alpha = (0.16, 0.04, 0.48, 0.32),\ r = 0.0$

Note on Domains and Uniqueness

The domains above are provided as examples only. Every negotiation uses different utility functions for both partners (different weights, value mappings, and reserved values), which means each negotiation is unique. To get a feel for how the competition works, try the online interface at: https://anac.cs.brown.edu/hanplay/app

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Alternating Offers Protocol (AOP)

The negotiation follows the **Stacked Alternating Offers Mechanism (SAOMechanism)**:

1. Negotiators take turns in a fixed order.
2. On each turn, the active negotiator observes the current offer on the table (if any) and selects one action:
    - **Reject & counter-offer**: Propose a new outcome $\omega \in \Omega$ (replaces the offer on the table).
    - **Accept**: Accept the current offer on the table, ending the negotiation in agreement.
    - **End**: Walk away, ending the negotiation in disagreement.
3. The negotiation is bounded by:
    - A maximum number of rounds N (e.g., 30–40 for HAN).
    - A wall-clock time limit T (e.g., 300 seconds for HAN).

If neither party reaches agreement or walks away before the bounds are exceeded, the negotiation ends by **timeout**. The strategy should not assume any specific value for U or Ω or T or N and can read them in real time during the negotiation.

All negotiators know the outcome-space and bounds (ω, N, T) and each agent i knows its own utility function U^i but not the opponent's utility function.

Offers

Each offer exchanged between negotiators consists of two optional components:

- **Text** (optional): A natural-language message delivered to the opponent (e.g., a persuasive argument or justification).
- **Outcome** (optional): A proposed outcome $\omega \in \Omega$, represented as a tuple of values matching the issue order.

An offer with an outcome places that outcome "on the table." An offer with only text removes the offer on the table (as usual) and places nothing to be accepted "on the table".

Agreement is only possible with an offer that contains an outcome — a negotiator can only accept when there is a concrete offer on the table.

Negotiation Termination

A negotiation ends in exactly one of three ways:

Termination	Condition	Payoff
Agreement	A negotiator accepts the current offer on the table (which must contain an outcome ω).	Both receive $U^j(\omega)$.
Leaving	A negotiator ends the negotiation voluntarily (walks away).	Both receive their reserved value r^j.
Timeout	The round limit N or time limit T is reached without agreement or walk-away.	Both receive their reserved value r^j.

Strategy

A **strategy** (also called an agent or negotiator) is a function that, at each round t, receives the current state of the negotiation and produces a decision. Formally, let H_t denote the negotiation history up to round t (all previous offers, texts, and decisions), let τ be the real time that passed since the start of the negotiation (which is bounded by T), let x_t be the (optional) text accompanying the offer from the partner, let $\omega_t \in \Omega \cup \{\emptyset\}$ denote the offer from the partner (offer on the table), and let TXT be any text. The strategy is a mapping:

$$S_t : (H_t,\ \omega_t, x_t,\ t, \tau;\ U^j) \longrightarrow \{\texttt{accept},\ \texttt{end}\} \cup (\Omega \cup \{\emptyset\} \times TXT)$$

At each round, the strategy observes the current offer and history, and returns one of:

- `accept` — accept the offer on the table ω_t (only valid when $\omega_t \neq \emptyset$)
- `end` — walk away from the negotiation
- an `offer` consisting of:
  - $\omega' \in \Omega \cup \{\emptyset\}$ — reject the current offer and propose a counter-offer $\omega' \in \Omega \cup \{\emptyset\}$ where $\emptyset$ means no counter offer.
  - $x' \in TXT$ — free text which can be empty.

The strategy has access to its own utility function U^j and reserved value r^j, but does **not** have access to the opponent's utility function.

In the actual implementation, the negotiator receives the latest offer only and can read the history on request or accumulate it itself.

Performance Metric

The primary metric is **advantage**, defined for negotiator j as:

$$\text{advantage}^j = A^j = \frac{U^j(\omega^*) - r^j}{\max_{\omega \in \Omega} U^j(\omega) - r^j}$$

where ω^* is the agreed outcome (or equivalently $A^j = 0$ in case of disagreement).

Tournament Mechanics

The competition is conducted in two stages:

Stage 1 — Qualification (Agent vs. Simulated Humans)

All submitted agents negotiate against **personality-adjusted LLM-based negotiators** that simulate human partners. The identity and implementation of these simulated partners are not revealed until after the competition concludes (though they are already used in the online competition during the submission period).

- **Score**: The mean advantage across all negotiations.
- **Fairness guarantee**: All agents face the same opponents in the same scenarios, ensuring comparable conditions.
- **Qualification**: The top scoring agents advance to the finals (exact number will be decided by the organizers).

Stage 2 — Finals (Agent vs. Real Humans)

Few top scoring agents from stage 1 participate in actual negotiations against human subjects (the exact number will be decided by the organizers).

Two evaluation criteria are used:

1. **Advantage-based score**: The same mean advantage metric as Stage 1.
2. **Subjective evaluation**: Human partners provide subjective ratings of the agents they negotiated with.